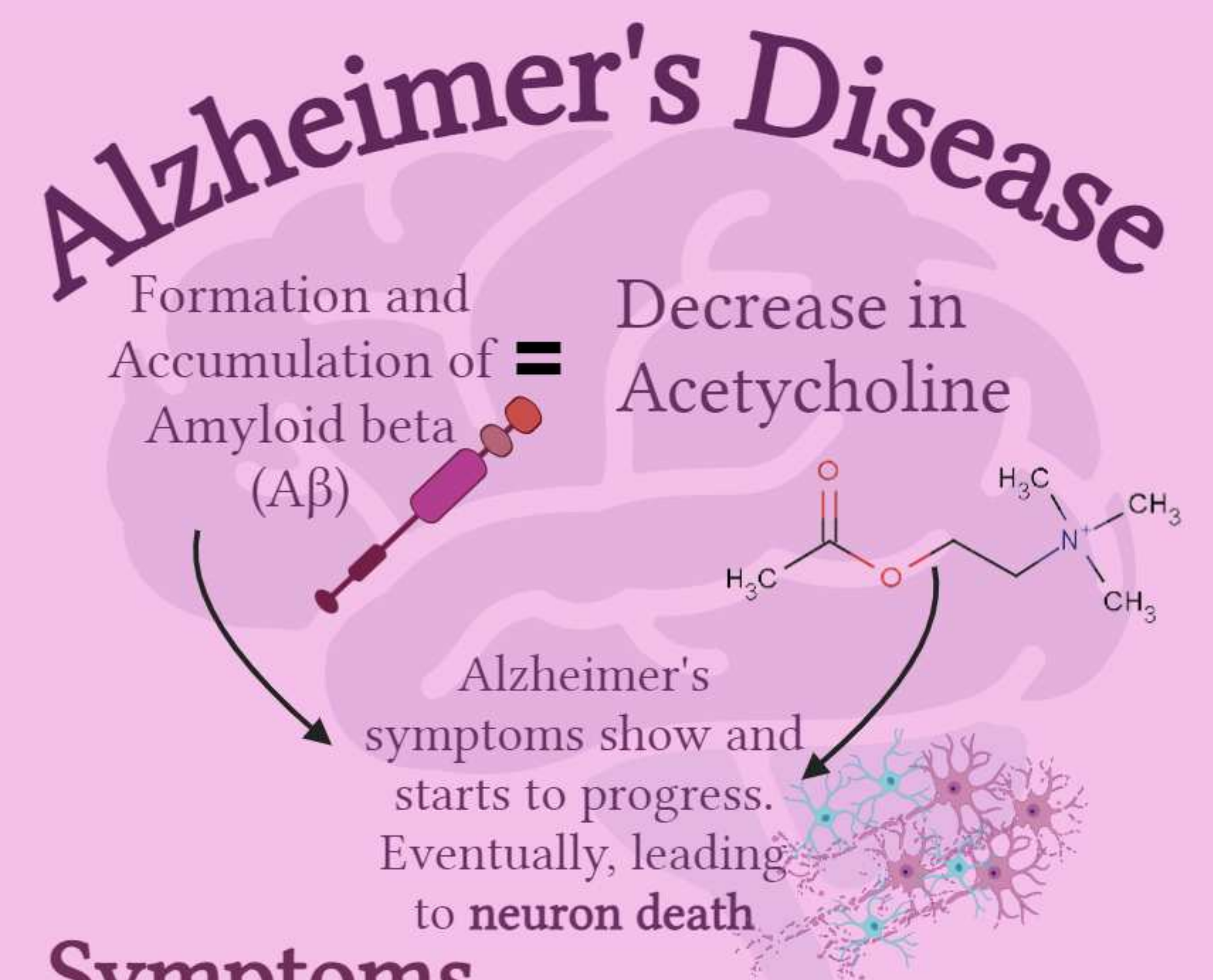


Determining if Epigallocatechin Gallate or Curcumin Works Better to Reduce Paralysis in *Caenorhabditis elegans* With Alzheimer's



Symptoms

- Struggling with daily tasks
- Confusion with time and place
- Difficulty solving problems
- Struggling to think

Most common in elders age 65 or above



Research Question

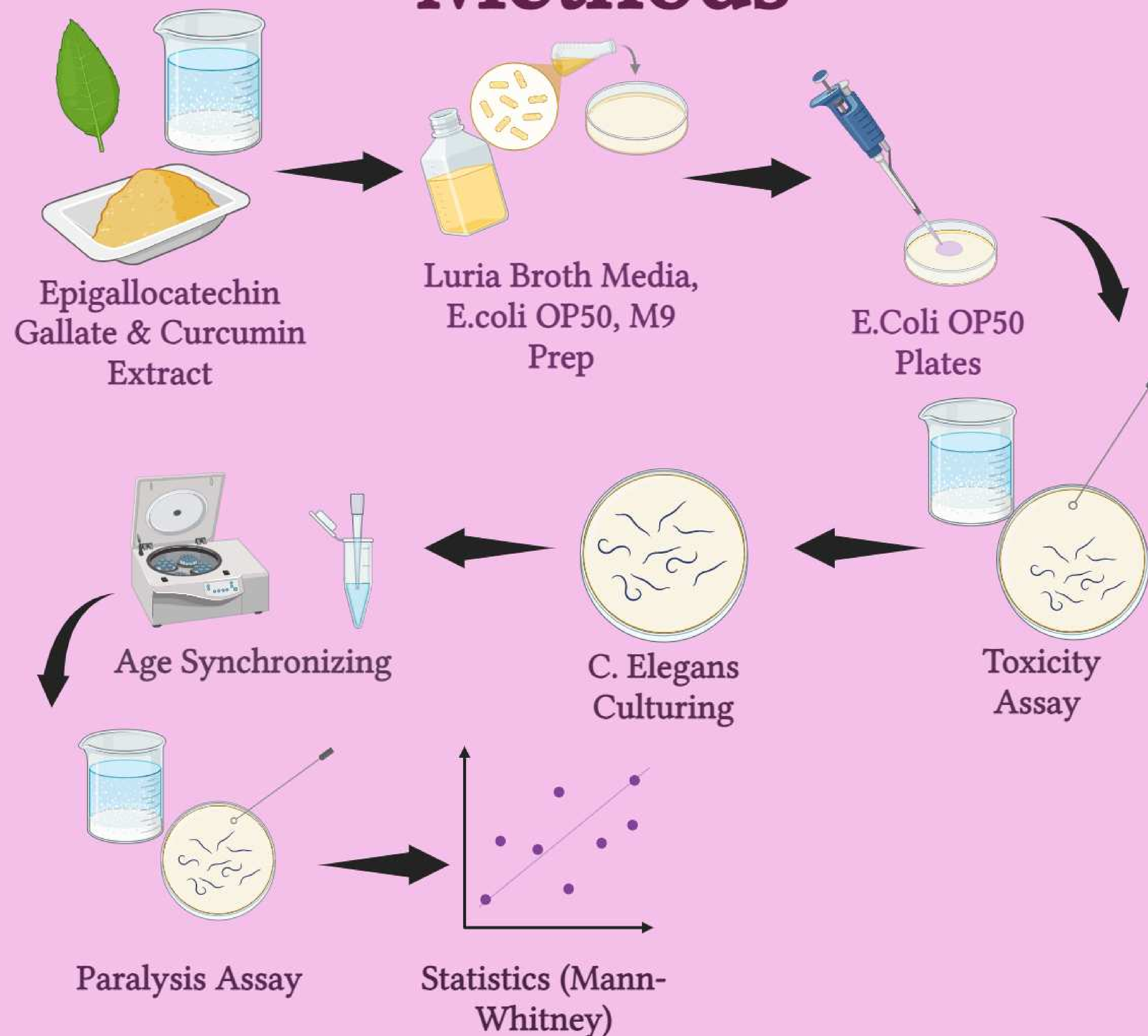
Which antioxidant, Epigallocatechin Gallate or Curcumin, will work better to improve locomotion in the transgenic Alzheimer's model *C. elegans*?

Hypothesis

If the worms in the CL4176 strain are treated with Epigallocatechin Gallate (green tea) or Curcumin, there will be a reduction in amyloid beta resulting in improved locomotion. This is because Epigallocatechin and Curcumin are both rich in polyphenols, which has been considered to be an inhibitor of amyloid beta. Yet, it is hypothesized that Epigallocatechin gallate will work more efficiently in reducing paralysis as there have been more studies conducted with EGCG compared to curcumin showing that it can be a potential cure for Alzheimer's.

IV: Epigallocatechin Gallate and Curcumin
DV: Paralysis Reduction
Control(-): Untreated CL4176 Worms

Methods



Toxicity Assay Data

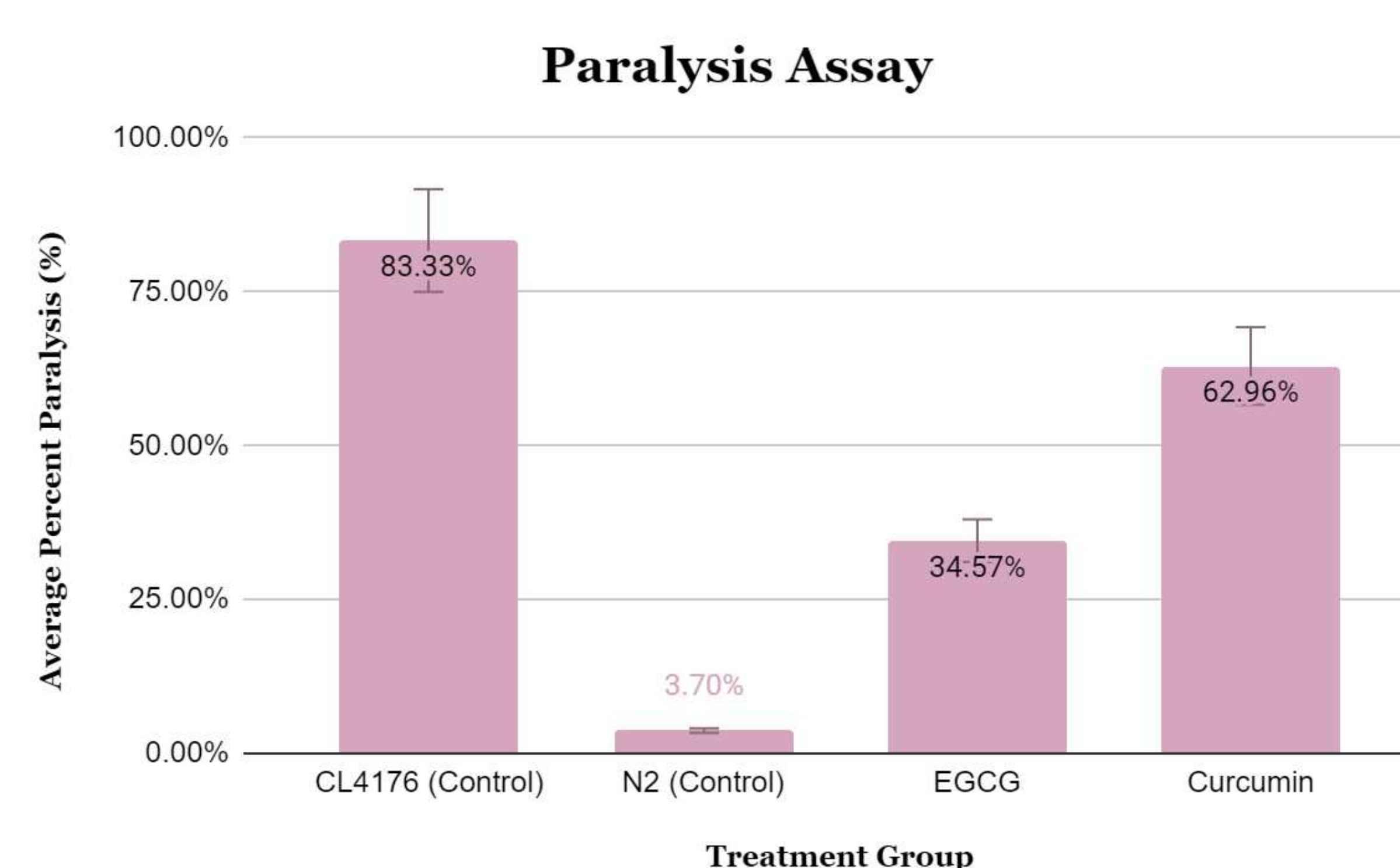
Table 1: Average Survival For Treatment Concentrations

Treatment Group	Average Survival (%)
EGCG 1x	100%
EGCG 0.5x	98.52%
EGCG 0.25x	97.78%
Curcumin 1x	91.67%
0.5x Curcumin	100%
0.25x curcumin	88.89%

Note. The data table above shows the best concentration of treatment for worms. EGCG 1x indicates 350 mg of Epigallocatechin Gallate. 0.5x Curcumin indicates 0.0875g of Curcumin

Paralysis Assay Data

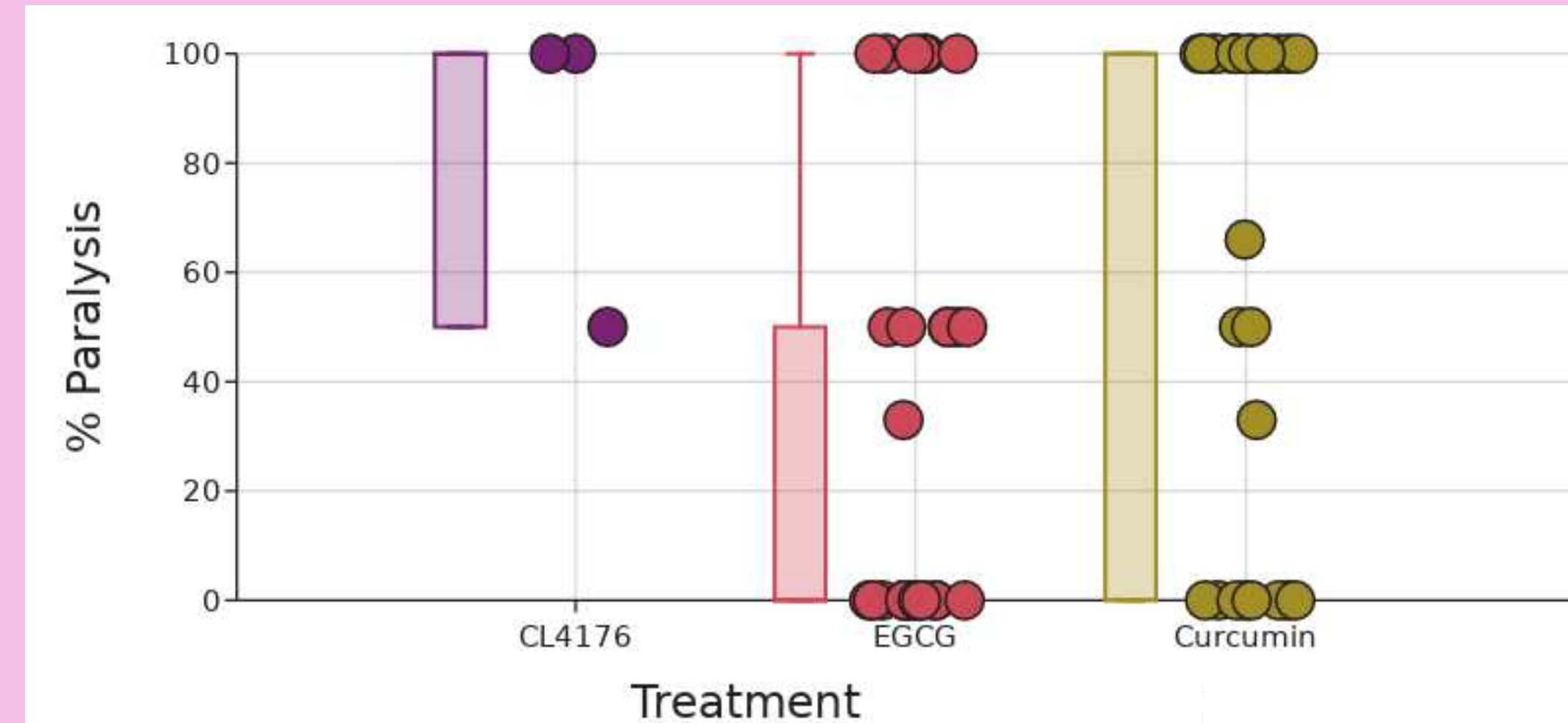
Figure 1: Average Paralysis For Treatment



Note. The Figure above shows the average percent of paralysis in each worm strain.

Discussion

Figure 2: Percent Paralysis in Treatments vs. Control



Note. The figure above shows median based graph of the treatments and control

Table 2: P-value for Each Treatment

	P-value
EGCG vs. CL4176	0.06
Curcumin vs. CL4176	0.54

Note. The table above shows the p-values for epigallocatechin gallate and Curcumin

- Hypothesis Supported** - Even though both reduced paralysis in worms, since the p-value of 0.06 is less than the p-value of 0.054, it can be concluded that Epigallocatechin Gallate works better in reducing the paralysis in the Alzheimer's Model Strain
 - Also shown in Figure 1, there was an overall of 34.57% paralysis in the worms treated with Epigallocatechin Gallate while there was an overall of 62.96% paralysis in the worms treated with Curcumin
- Epigallocatechin Gallate can help be a potential cure for Alzheimer's disease
- Future works for this project can be investigating the effects of the combination treatments of these two and using gene expression analysis to investigate how Epigallocatechin Gallate affects oxidative stress or the neurodegenerative pathways in *c elegans*.

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